# ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

**ORGANISATION OF ISLAMIC COOPERATION (OIC)**

**Department of Computer Science and Engineering (CSE)**

**QUIZ 3** **SUMMER SEMESTER, 2023-2024**

**DURATION: 25 MINUTES** **FULL MARKS: 20**

# CSE 4411: Data Communication and Networking

**Programmable calculators are not allowed. Do not write anything on the question paper.**

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

STUDENT ID:

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1. For the generator G (= 1001), answer the following questions. 2 + 1 + 3 (CO3) (PO2)

   a) Why can it detect any single bit error in data D?

   **Solution:**
   Because if any single bit is flipped the remainder will not be zero anymore

   b) Can the above G detect any odd number of bit errors? Why?

   **Solution:**
   Yes, the key insight here is that G can be divided by 11 (3) as $(x^3+1) = (x+1)(x^2-x+1)$, but any number of odd-number of 1s cannot be divided by 11. Thus, a sequence (not necessarily contiguous) of odd-number bit errors cannot be divided by 11, thus it cannot be divided by G.

2. Consider the network shown below. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Suppose eBGP and iBGP are used for the inter-AS routing protocol. Initially suppose there is no physical link between AS2 and AS4. $4 \times 1$ (CO3) (PO2)

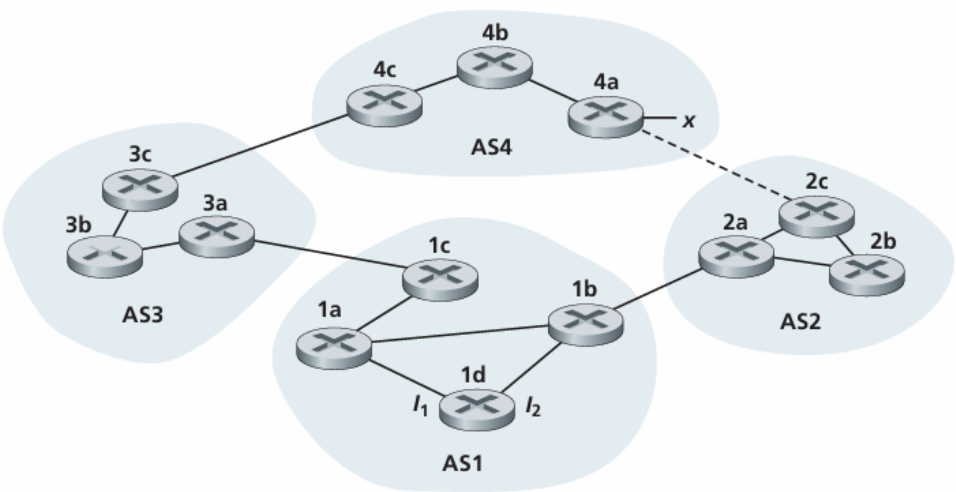


**Figure 1:** Topology for Question 2

   a) Router 3c learns about prefix x from which routing protocol: OSPF, RIP, eBGP, or iBGP?

**Solution:**

eBGP

b) Router 3a learns about x from which routing protocol?

**Solution:**

iBGP

c) Router 1c learns about x from which routing protocol?

**Solution:**

eBGP

d) Router 1d learns about x from which routing protocol?

**Solution:**

iBGP

3. Answer true or false, with one line of justification. $2 \times 2$ (CO3) (PO2)

a) If all the links in the Internet were to provide reliable delivery service, the TCP reliable delivery service be redundant.

**Solution:**

False, while individual links might guarantee reliable delivery, TCP ensures that data is delivered correctly and in order from the source to the destination across *multiple links*.

b) The two-dimensional parity checks can correct a two bits error

**Solution:**

False, 2-D parity can detect many two-bit errors but cannot generally correct them, because row/column parity syndromes do not uniquely locate two flipped bits (they can cancel or be ambiguous).

4. Prove that the efficiency of slotted Aloha is around 37%. 6 (CO3) (PO2)
Given,

$$\lim_{N\to\infty}\left(1-\frac{1}{N}\right)^{N} = e^{-1}$$

**Solution:**

For $N$ nodes, each transmitting in a slot with probability $p$, the probability that *exactly one* node transmits (the efficiency) is

$$E(p) = Np(1-p)^{N-1}.$$

Find $p^*$ that maximizes $E(p)$, compute the maximum $E(p^*)$, and show the limit as $N \to \infty$ gives $1/e$.

**First derivative and optimum (3 Marks)**

Differentiate $E(p)$ with respect to $p$. Using product and chain rules,

$$E(p) = Np(1-p)^{N-1},$$

$$\frac{dE}{dp} = N\Big[(1-p)^{N-1} + p\,(N-1)(1-p)^{N-2}(-1)\Big]$$

$$= N(1-p)^{N-2}\Big[(1-p) - p(N-1)\Big]$$

$$= N(1-p)^{N-2}(1-pN).$$

Set $\frac{dE}{dp} = 0$. Since $(1-p)^{N-2} = 0$ corresponds to the trivial boundary $p = 1$, the interior critical point satisfies

$$1 - pN = 0 \quad \Rightarrow \quad \boxed{p^* = \frac{1}{N}.}$$

**Maximum efficiency (3 Marks)**

Substitute $p^* = 1/N$ into $E(p)$:

$$E(p^*) = N \cdot \frac{1}{N}\left(1 - \frac{1}{N}\right)^{N-1} = \left(1 - \frac{1}{N}\right)^{N-1}.$$

*Limit as N → ∞* We use the standard exponential limit. First recall

$$\lim_{N\to\infty}\left(1 - \frac{1}{N}\right)^{N} = e^{-1}.$$

Then

$$\lim_{N\to\infty} E(p^*) = \lim_{N\to\infty}\left(1 - \frac{1}{N}\right)^{N-1} = \lim_{N\to\infty}\frac{\left(1-\frac{1}{N}\right)^{N}}{1-\frac{1}{N}} = \frac{1/e}{1} = \frac{1}{e}.$$

(Equivalently, take natural logs: $\ln\left((1-\frac{1}{N})^N\right) = N\ln(1-\frac{1}{N}) \to -1$ by the Taylor expansion $\ln(1-x) = -x + o(x)$ as $x \to 0$.)

**Final boxed result**

$$\boxed{p^* = \frac{1}{N}, \qquad \max_p E(p) = \left(1 - \frac{1}{N}\right)^{N-1}, \qquad \lim_{N\to\infty}\max_p E(p) = \frac{1}{e} \approx 0.3679}$$